ModelIt!

Teacher's Guide

# Climate Feedback Loops: Why Small Changes Get Big

How $CO_2$, Temperature, Ice, and Reflectivity Create a Runaway System

## Quick Reference

| | |
|---|---|
| **Grade Level** | Grade 08L2 |
| **NGSS Standard** | MS-ESS2-6, MS-ESS3-5 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |

ModelIt! — Discovery Collective

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# 1 NGSS Standards & CAST Alignment

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**Performance Expectation: MS-ESS2-6, MS-ESS3-5**

Develop and use a model to describe how unequal heating and rotation of the Earth cause patterns of atmospheric and oceanic circulation that determine regional climates. Ask questions to clarify evidence of the factors that have caused the rise in global temperatures over the past century.

## Three-Dimensional Learning

| | |
|---|---|
| **Science Practice** | **Developing and Using Models** — Students develop a model demonstrating how feedback loops in Earth's climate system amplify initial changes. |
| **Disciplinary Core Idea** | **ESS3.D Global Climate Change** — Human activities are major factors in global warming. Rising temperatures create feedback loops that can accelerate change beyond initial causes. |
| **Crosscutting Concept** | **Stability and Change** — Students analyze how feedback loops can push a system past a stability threshold (tipping point) into a fundamentally different state. |

## CAST Testing Alignment

**MULTIPLE CHOICE:**

Arctic ice cover has decreased by 40% since 1980. A student claims this creates a feedback loop that causes more warming. Which chain of events supports this claim?

**CONSTRUCTED RESPONSE:**

Using evidence from your model, explain why a small increase in $CO_2$ emissions can lead to a disproportionately large increase in global temperature. Include the role of albedo and the feedback loop in your explanation.

# 2 Scoring Rubric

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### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

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**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Earth's climate is regulated by complex feedback loops that can either stabilize or destabilize the system. Understanding these feedback mechanisms is essential for predicting how human-caused CO2 emissions will affect global climate in the coming decades.

## The Greenhouse Effect

Earth's atmosphere contains gases (CO2, methane, water vapor) that trap outgoing infrared radiation. This greenhouse effect keeps Earth approximately 33°C warmer than it would be without an atmosphere. Human activities have increased atmospheric CO2 from 280 ppm (pre-industrial) to over 420 ppm today — a 50% increase.

## The Ice-Albedo Feedback

Ice and snow reflect 80-90% of incoming sunlight (high albedo). Open ocean and dark land absorb 90-94% of sunlight (low albedo). As warming melts ice, the exposed dark surface absorbs more heat, causing more warming, which melts more ice. This is the ice-albedo positive feedback loop — the most significant amplifying feedback in Earth's climate system.

## Climate Tipping Points

Scientists have identified several potential tipping points: complete Arctic summer ice loss, Greenland ice sheet collapse, permafrost thaw releasing stored methane, Amazon rainforest dieback, and Atlantic circulation shutdown. Once crossed, these tipping points may be irreversible on human timescales. Current estimates suggest some tipping points could be crossed between 1.5°C and 2°C of warming above pre-industrial levels.

## The Carbon Budget

Scientists estimate humanity can emit approximately 400 billion more tons of CO2 while maintaining a 50% chance of staying below 1.5°C of warming. At current emission rates (approximately 40 billion tons per year), this budget will be exhausted within a decade. The feedback loops in the climate system mean that every fraction of a degree matters.

# 4 The LEVER Framework

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### About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L** **Locate the System**
Students identify $CO_2$ emissions rate, atmospheric $CO_2$ level, global temperature, ice cover, and albedo as system components.

**E** **Establish Relationships**
Students determine that emissions drive $CO_2$ levels, $CO_2$ drives temperature, temperature reduces ice, ice affects albedo, and albedo feeds back to temperature — creating a loop.

**V** **Visualize & Model**
Students build the climate feedback model showing how a linear cause-effect chain becomes a self-reinforcing loop.

**E** **Evaluate Outcomes**
Students compare pre-industrial, current, and runaway scenarios to see how the feedback loop amplifies with higher emissions.

**R** **Revise & Extend**
Students add permafrost thaw or ocean heat absorption to model additional feedback mechanisms.

## Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

| | |
|---|---|
| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes. |
| **As a "what-if" tool** | When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

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## Slide 1: Cover

2 min

**Say:** *"This glacier is 10,000 years old. It's disappearing in decades. Why?"*

**Watch For**

Show time-lapse video of glacier retreat. Let the visual create urgency.

## Slide 2: Learning Objectives

3 min

**Say:** *"Today we model why Earth's climate has a hidden 'accelerator pedal' built in."*

**Watch For**

Pre-teach 'feedback loop' with an audio feedback analogy (microphone near speaker).

## Slide 3: Big Question

3 min

**Say:** *"A little CO2 shouldn't cause this much change. So why does it?"*

**Watch For**

Think-pair-share: What's the difference between a gradual change and a snowball effect?

## Slide 4: LEVER Framework

2 min

**Say:** *"We'll build a model that shows how Earth's climate amplifies small changes."*

**Watch For**

Preview the concept of feedback loops — this is the first loop model in the curriculum.

## Slide 5: Activity 1: Components

8 min

**Say:** *"What are the key parts of the climate feedback system?"*

**Watch For**

Guide sorting. Emphasize that most components are INTERNAL — only emissions are truly external.

## Slide 6: Activity 2: Connections

8 min

**Say:** *"Watch: the last arrow connects BACK to the beginning. That's what makes it a loop."*

**Watch For**

Draw the loop step by step. The moment students see the loop close is the 'aha' moment.

## Slide 7: Activity 3: Simulation

10 min

**Say:** *"At low emissions, the loop barely runs. At high emissions, watch what happens."*

**Watch For**

Students predict, then run all three scenarios. Focus on the acceleration in the runaway scenario.

## Slide 8: Discoveries

5 min

**Say:** *"This is why scientists talk about tipping points. Small changes can become unstoppable."*

**Watch For**

Connect to real Arctic ice data. Discuss what 'irreversible' means.

## Slide 9: STEM Challenge

12 min

**Say:** *"Your city council wants a plan. Where in the loop do you intervene? Show your evidence."*

**Watch For**

Teams design action plans targeting specific loop components. Present with data.

# 6 Answer Key & Expected Responses

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## Component Sorting

- **EXTERNAL:** CO2 Emissions Rate
- **INTERNAL:** Atmospheric CO2 Level, Global Temperature, Ice Cover, Albedo (Reflectivity)

## Relationships

- **CO2 Emissions Rate to Atmospheric CO2 Level:** POSITIVE (+) — More fossil fuel burning releases more CO2 into the atmosphere, increasing the concentration of greenhouse gases.
- **Atmospheric CO2 Level to Global Temperature:** POSITIVE (+) — Higher CO2 concentrations trap more outgoing heat through the greenhouse effect, raising Earth's average temperature.
- **Global Temperature to Ice Cover:** NEGATIVE (-) — Higher temperatures melt ice sheets, glaciers, and sea ice, reducing the total area of frozen surfaces on Earth.
- **Ice Cover to Albedo:** POSITIVE (+) — More ice means more reflective surface area to bounce sunlight back to space. Less ice means lower reflectivity.
- **Albedo to Global Temperature:** NEGATIVE (-) — Higher albedo reflects more sunlight, cooling Earth. Lower albedo absorbs more sunlight, warming Earth. This is where the FEEDBACK LOOP closes — lower albedo causes more warming, which melts more ice, which lowers albedo further.

## Key Discoveries

- CO2 traps heat through the greenhouse effect, raising global temperatures
- Higher temperatures melt ice, which reduces albedo (reflectivity)
- Lower albedo means more sunlight absorbed instead of reflected, causing MORE warming
- This creates a positive feedback loop: warming → ice melt → lower albedo → more warming → more ice melt
- Feedback loops explain why small CO2 increases can trigger disproportionately large temperature changes

### Multiple Choice Post-Assessment Answer Key

**1. Your model shows that at 90% CO2 Emissions Rate, Global Temperature increases exponentially rather than linearly. Which model feature explains this acceleration?**

**Correct: B) The positive feedback loop where warming reduces ice, which lowers albedo, which causes more warming, amplifies the initial CO2 effect**

*Correct! The feedback loop amplifies the initial warming. CO2 raises temperature, which melts ice, which reduces albedo (reflectivity), which causes Earth to absorb more sunlight, which raises temperature further. Each cycle through the loop adds more warming beyond what CO2 alone would cause.*

**2. In the model, Albedo has a NEGATIVE relationship with Global Temperature. A student is confused because lower albedo causes higher temperature, which seems like it should be positive. Which explanation correctly resolves this?**

**Correct: B) Higher albedo reflects more sunlight away from Earth, which cools the surface. The negative sign means the two values move in opposite directions.**

*Correct! In system models, a negative relationship means the values move in opposite directions. Higher albedo means more reflection and LOWER temperature. Lower albedo means more absorption and HIGHER temperature. The negative sign correctly describes this inverse relationship.*

**3. The model shows that even after $CO_2$ Emissions Rate is reduced to 0%, Global Temperature continues rising for several decades. What does this evidence reveal about the climate system?**

**Correct: B) The feedback loop has its own momentum. Reduced ice cover continues absorbing extra sunlight, maintaining warming even without new $CO_2$ input**

*Correct! The feedback loop has inertia. Even without new $CO_2$, the already-reduced ice cover continues absorbing more sunlight than ice would, maintaining warming. The system needs time to re-freeze ice and restore albedo. This is why prevention is easier than reversal.*

**4. A student claims that the ice-albedo feedback loop is a 'positive feedback' that produces positive results for Earth. Using the model, which response best corrects this misunderstanding?**

**Correct: B) In systems science, 'positive' means the feedback amplifies the original change, not that the outcome is good. The ice-albedo loop amplifies warming, which has harmful consequences.**

*Correct! In systems science, 'positive' describes the direction of amplification, not the desirability of the outcome. A positive feedback loop pushes change further in the same direction. The ice-albedo loop amplifies warming, making it worse, even though the word 'positive' might suggest something good.*

**5. Using your model, you must advise a city council on climate action. The model shows that intervening at $CO_2$ Emissions Rate has the largest effect on all downstream components. Why is this the most effective intervention point?**

**Correct: B) Because $CO_2$ Emissions Rate is the external input that drives the entire feedback loop. Reducing it lowers Atmospheric $CO_2$, which slows temperature rise, which preserves ice, which maintains albedo.**

*Correct! $CO_2$ Emissions Rate is the only external input in the model. All other components are internal and respond to each other. Reducing the external input slows the entire chain: less $CO_2$ means less greenhouse warming, which means less ice melt, which preserves albedo. Targeting the input is more effective than treating downstream symptoms.*

# 7 Collaborative Learning Protocols

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### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
|---|---|
| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
|---|---|
| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

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### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

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### Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
|---|---|
| "Climate change is just gradual warming" | Feedback loops mean climate change is NOT gradual — it can accelerate. The ice-albedo feedback creates a snowball effect where warming causes changes that cause MORE warming. The relationship between CO2 and temperature is not a straight line but a curve that steepens over time. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Melting ice is the only problem" | Ice loss matters not primarily because of the ice itself, but because of what it reveals. Dark ocean and land absorb vastly more sunlight than ice, creating additional warming. The loss of reflective surface is a planetary thermostat being broken. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "We can just stop emissions and everything will be fine" | Even if emissions stopped today, the feedback loops already in motion would continue for decades. Reduced ice cover continues absorbing extra heat. Warming oceans continue releasing stored heat. The climate system has inertia — like a freight train, it doesn't stop the moment you release the brake. |